AUTONOMOUS SOFTWARE EXPERIMENTS ON HERA: ANNEX B - DATAPOOL

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The mission data pool is a central set of observable spacecraft parameters maintained by Hera's on-board software. It includes data coming from different units and subsystems.

For the Experimental Software (ESW) running on Core 1, data pool access is read-only. The ESW can read a selected set of spacecraft state parameters, but it cannot command actuators through the data pool. The Spacecraft Data Pool is accessed through the Data Pool read functions (e.g. `Hera_Read_Parameter_uint32`, `Hera_Read_Parameter_float64`, etc.). These functions return the current value of a specific parameter identified by a unique parameter reference (`parameter_ref`).

The full list of data pool parameters will be provided to the selected experiments.

1. INPUT PARAMETERS FOR THE EXPERIMENT SW

A limited number of data pool parameters will be reserved for the Experiment SW (maximum 10–15). Ground control can use these parameters to provide configuration to the Experiment SW before each execution window. Ground will set these parameter values only once per execution window, immediately before the experiment starts.

2. AOCS / GNC (GUIDANCE, NAVIGATION & CONTROL)

Parameters used to determine attitude/rates and support spacecraft control (including sensor/actuator selection and validity). This includes both raw telemetry and processed outputs:

- Modes, transitions, and counters (current mode/submode and time in mode)
- Validity flags and selection arbitration (which sensor/actuator is valid and selected)
- Gyro measurements, time tags, and health
- Star tracker telemetry and decoded status
- Sun sensor / coarse sun sensor values and selection
- Reaction wheel data (speed, torque, friction terms, momentum estimates)
- Estimation outputs (attitude/rate/bias estimates)

3. DATA HANDLING & MEMORY (MMU)

Parameters used to monitor mass memory and related functions:

- Router configuration/status (routing and interface diagnostics)
- Memory statistics and health counters
- Storage/downlink selection status

- File delivery (CFDP) tracking parameters

4. POWER & THERMAL (PCDU / THER)

Parameters used to monitor power distribution and thermal control:

- PCDU line switching/LCL status (currents, voltages, protections)
- TM/TC lane selection and status
- Bus/interface selection related to PCDU
- Heater group selection/activation/state
- Temperature measurements (per unit and sensor point)

5. AVIONICS INTERFACES AND BUSES (SPW / RTU / 1553)

Parameters used to monitor main data links and interface chains on the spacecraft:

- SpaceWire link status (link state for endpoints and general link status variables)
- SpaceWire error counters (e.g. RMAP/CRC-related error counters)
- RTU acquisition status and readback (validity of acquisitions and command/status read-back)
- RTU acquired signals (engineering values and on-board calibrated values)
- Routing / path selection parameters (e.g. PATH parameters used to select payload/data paths)

6. EQUIPMENT MANAGEMENT (SELECTION, COMM STATUS)

Equipment manager view of the spacecraft: which units are selected, what would be selected in SAFE modes, and the overall unit state and communication status:

- Nominal selection tables (which unit is selected in nominal mode, A/B selection for key equipment)
- SAFE / SURVIVAL selection tables (the equivalent selections used in SAFE/SURVIVAL behaviour)
- Unit communication status (whether a unit is reachable / not reachable)
- Assembly activation state (activation/state per equipment group)

7. TM/TC AND BANDWIDTH

Telemetry throughput, bandwidth allocation, and monitoring of telecommand reception:

- Bandwidth allocation and PTME status (e.g. allocation of slots/virtual channels and PTME status)
- TM loss counters (e.g. counters for lost TM packets due to lack of bandwidth)
- TC link monitoring and recovery (e.g. time since last TC reception, recovery enable flags, and recovery counters)

8. PROPULSION (CPS)

CPS unit selection and propulsion-related telemetry:

- CPS unit selection (A/B unit selection parameters)
- CPS communication status (where applicable)
- CPS telemetry for trending (e.g. cumulative on-times, temperatures, and other long-term monitoring metrics)

9. PALT PAYLOAD

PALT payload configuration and status:

- PALT configuration and status (instrument settings and measurement-related parameters)